

Research protocol (frozen) and change record

source: research/protocol_v1.json + protocol_changes.csv — v1.4.0

```
{
  "protocol_version": "1.1",
  "frozen_at": "2023-10-20",
  "project_id": "XYJ-LAB-BILLING-RESEARCH",
  "data_mode": "DEMO",
  "research_questions": {
    "RQ1": "Does uncertainty-preserving event reconstruction reduce wrong links and control manual review volume vs simple identifier joins and appropriate existing methods?",
    "RQ2": "Does adding record-quality and processing-stage signals improve within-horizon first-event prediction vs aging and basic baselines?"
  },
  "target_definition": {
    "name": "FIRST_OBSERVED_POSITIVE_ALLOCATION_OR_NO_PAYMENT_CLOSURE",
    "status": "DEMO_ASSUMPTION_REAL_CONFIRMATION_REQUIRED",
    "landmark_days": 7,
    "horizon_days": 30,
    "interval_days": 5,
    "outcomes": [
      "P_payment_by_H",
      "P_close_no_payment_by_H",
      "P_still_open_by_H"
    ],
    "censoring": "STOP_AT_RELIABLE_OBSERVATION_END",
    "insufficient_data_action": "ABSTAIN"
  },
  "linkage_methods": [
    "exact_id",
    "constrained_linkage"
  ],
  "models": [
    "B0_age_stage_baseline",
    "B1_discrete_multinomial",
    "C1_quality_augmented"
  ],
  "split": {
    "policy": "TEMPORAL_GROUPED_BY_LINEAGE",
    "demo_ratios": [
      0.6,
      0.2,
      0.2
    ],
    "lock_test_set": true
  },
  "metrics": {
    "module_a": [
      "precision",
      "recall",
      "wrong_merge_count",
      "wrong_split_count",
      "refusal_ratio",
      "coverage"
    ]
  }
}
```

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    ],
    "module_b": [
        "log_loss",
        "brier_three_class",
        "grouped_calibration",
        "class_counts",
        "refusal_ratio",
        "censoring_ratio"
    ]
},
"conclusion_conditions": {
    "positive": "Candidate improves a pre-specified metric on the locked test set beyond a pre-specified margin, with uncertainty intervals excluding null.",
    "negative": "No improvement, or worse. Reported as-is; not repaired by changing ground truth.",
    "inconclusive": "Insufficient sample or class counts; INSUFFICIENT_DATA recorded, no probabilities fabricated."
},
"feature_whitelist_frozen": true,
"notes": "DEMO results support engineering validation only. Research claims require REAL-LOCAL evaluation.",
"updated_at": "2024-08-06",
"change_history": [
    {
        "change_id": "PC01",
        "date": "2024-08-06",
        "summary": "interval_days 1->5 after sparsity findings; see research/protocol_changes.csv"
    }
]
}

```

Change record

change_id	date	affected_section	change	reason	evidence	decision_by	status
PC01	2024-08-06	protocol section Module B interval	interval_days 1 -> 5 (H=30 becomes 6 intervals)	Event sparsity at 1-day granularity made interval-level conditional probabilities numerically unstable (payment events concentrate around 10-30 days); 5-day intervals raise per-interval event rates to a learnable range while preserving within-horizon discrimination	Initial module B run 2024-08-05 (interval=1): B1 snap log-loss 11.3 vs B0 0.74; interval-level none share >95%	Applicant	applied (protocol v1.1)